DBCTNet: Double Branch Convolution-Transformer Network for Hyperspectral Image Classification

Rui Xu, Xue-Mei Dong, Weijie Li, Jiangtao Peng, *Senior Member, IEEE*, Weiwei Sun, *Senior Member, IEEE*, and Yi Xu

***Abstract*— Currently, deep learning (DL) methods represented by convolutional neural networks (CNNs) or Transformers are of great interest in hyperspectral image (HSI) classification. And recent works show that hybrid models using CNN and Transformer modules are expected to achieve better performance than when they are used alone. However, these hybrid models applied to HSI classification consider the combination of 2-D CNN and Transformer, which makes the models have high computational complexity. And the information of multiple spectral dimensions different from ordinary RGB images has not been fully excavated. Based on this, we propose, a double branch Convolution-Transformer network (DBCTNet). Specifically, a MSpeFE module is used for multiscale spectral feature extraction at the early stage of the proposed network. Then, a ConvTE block is designed to improve the original Transformer encoder (TE), where a Conv spectral projection unit and a convolutional multihead self-attention (CMHSA) unit are proposed to extract spatial and global spectral features. A double branch module is further built based on 3-D CNN and ConvTE. This module can fully integrate spatial and local–global spectral features, while also having low computational complexity. Experiment results on four public datasets, Pavia University, Houston, WHU-Hi-LongKou, and HuangHeKou, show that DBCTNet achieves satisfactory performance with a small number of parameters and relatively excellent efficiency compared to nine other networks. The implement of DBCTNet will be available publicly at (https://github.com/xurui-joei/DBCTNet).**

***Index Terms*— Convolutional neural networks (CNNs), hyperspectral image (HIS) classification, multiscale, Transformer.**

Manuscript received 14 October 2023; revised 19 December 2023, 16 January 2024, and 17 February 2024; accepted 18 February 2024. Date of publication 20 February 2024; date of current version 28 February 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 42171351 and Grant 42122009, in part by the Characteristic and Preponderant Discipline of Key Construction Universities in Zhejiang Province (Zhejiang Gongshang University-Statistics) and Collaborative Innovation Center of Statistical Data Engineering Technology and Application, and in part by the "Digital+" Discipline Construction Management Project of Zhejiang Gongshang University under Grant SZJ2022C011. *(Corresponding author: Xue-Mei Dong.)*

Rui Xu, Xue-Mei Dong, and Weijie Li are with the School of Statistics and Mathematics, Zhejiang Gongshang University, Hangzhou 310018, China (e-mail: dongxuemei@zjgsu.edu.cn).

Jiangtao Peng is with the Hubei Key Laboratory of Applied Mathematics, Faculty of Mathematics and Statistics, Hubei University, Wuhan 430062, China (e-mail: pengjt1982@hubu.edu.cn).

Weiwei Sun is with the Department of Geography and Spatial Information Techniques, Ningbo University, Ningbo 315211, China (e-mail: sunweiwei@nbu.edu.cn).

Yi Xu is with the School of Control Science and Engineering, Dalian University of Technology, Dalian 116024, China (e-mail: yxu@dlut.edu.cn).

Digital Object Identifier 10.1109/TGRS.2024.3368141

I. Introduction

With the help of hyperspectral sensors, hyperspectral images (HSIs) contain rich spectral and spatial information of ground objects through hundreds of spectral narrow bands. In recent years, HSIs have attracted the attention of researchers in many fields and have been applied to geological exploration [1], anomaly detection [2], [3], [4], agriculture [5], atmospheric sciences [6], and military reconnaissance [7].

As one of important tasks of HSI applications, HSI classification has attracted the interest of many researchers. Based on spectral and spatial information, a lot of conventional HSI classification methods often utilize linear regression [8], support vector machines [9], maximum a posteriori estimator [10], sparse representation models [11], and spectral clustering [12] to output classification maps. Most of them require manual features dependent on domain knowledge [13].

Recently, deep learning (DL) methods have provided new solutions for HSI classification. Recurrent neural networks (RNNs) [14], autoencoders [15], [16], generative adversarial networks (GANs) [17], graph neural networks (GNNs) [18], [19], [20], involution [21], and convolutional neural networks (CNNs) have been used to design HSI classification algorithms. Being a quite effective method, CNNs have become the mainstream DL method for HSI tasks [22], [23], [24], [25]. The methods based on 2-D CNNs overcome the shortcomings of conventional methods by automatic feature extraction. But HSIs usually contain hundreds of spectral bands, which leads to an increase in the number of parameters for these algorithms. Meng et al. [26] designed a residual dense asymmetric convolutional network (RDACN), replacing the ordinary square convolutional kernels with the asymmetric convolutional kernels, which can reduce CNN parameters. Subsequently, 3-D CNNs were applied in HSI classification. Zhong et al. [27] proposed SSRN, an end-to-end 3-D CNN with spectral and spatial residual blocks to learn discriminative features from spectral and spatial information, and utilized batch normalization to improve the performance of models. HybridSN, developed by Roy et al. [28], extracted joint spatial-spectral features with 3-D CNN and further extracted abstract-level spatial features with 2-D CNN. Roy et al. [29] further described A2S2KResNet, which learned selective 3-D convolutional kernels with an adaptive spectral-spatial kernel based on the attention mechanism and used modified 3-D residual blocks to extract joint spatial-spectral features. Although 3-D convolution kernels

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:32:34 UTC from IEEE Xplore. Restrictions apply.

need fewer parameters than 2-D convolution kernels, these 3-D CNN-based methods still have a relatively large number of parameters. LiteDepthwiseNet, recently proposed by Cui et al. [30], significantly reduced the number of parameters and floating-point operations (FLOPs) with depthwise separable convolution (DSConv) and achieved remarkably excellent accuracy. However, CNNs are expert at modeling local features and spatial features instead of global features or long-distance dependencies [31], [32], due to the limited receptive fields.

With Transformers making a huge impact on the field of natural language processing (NLP) [33], [34], Dosovitskiy et al. [35] first applied Vision Transformer (ViT) in computer vision tasks and achieved better performance than CNNs. Due to its excellent ability to exploit global information and long-distance dependencies, a series of Transformer-based models following ViT showed pleasing competitiveness in image processing [36], [37], [38]. However, Transformer-based methods usually have a large number of parameters and FLOPs [39], [40]. Furthermore, due to the lack of inductive bias suitable for images, such as translation invariance [41], Transformer-based methods generally are insensitive to local information [42] and require more training data and training time [43]. Therefore, many researchers combine CNNs with Transformers to construct models which have the advantages of both methods. These models have been shown to improve performance, speed up the training process, and are lightweight compared to Transformers [40], [41], [42], [44], [45].

There has also been some recent Transformer-based works for HSI classification tasks. Hong et al. [46] proposed SpectralFormer, which used Transformers to mine the sequence attributes of spectral signatures and designed a cross-layer skip connection to retain more valuable information in the layer-wise propagation. Tu et al. [47] proposed LSFAT, a local semantic feature aggregation-based transformer which represents long-range dependencies of multiscale features more efficiently. Yang et al. [48] proposed HiT, which embedded convolution operations into the Transformer encoder (TE) to extract spectral features and the local spatial features. Roy et al. [49] proposed morphFormer, which used spectral and spatial morphological convolution operations to improve the interaction between the structural and shape information of different tokens. Zhang et al. [50] developed CTMixer, which constructed a double-branch structure based on 2-D CNN and a TE to extract local–global features. Ouyang et al. [51] proposed the HybridFormer, which exploited the local representation ability of 2-D CNN to embed multiple granularity tokens and replace multihead self-attention (MHSA) with spatial-spectral attention to capture the global dependencies between different tokens. Jiang et al. [52] integrated graph generative structure into the Transformer architecture to capture local-to-global correlations, which generated excellent classification results for HSIs.

Although these existing hybrid methods performed well, several problematic issues remain. CTMixer [50], drawing inspiration from ViT, serializes spatial pixels for attention calculation, without fully utilizing the global information capability of the Transformer in view of the high spectral dimension characteristic of HSI data. In addition, CTMixer combines 2-D CNN with Transformers in parallel, lacking the utilization of local spectral information. HybridFormer [51] integrates 2-D CNN with Transformers, sequentially executing spatial and spectral attentions for feature extraction. But a serial combination of convolution and self-attention cannot run through the entire network architecture, limiting the continuous modeling of local and global representations [53]. Furthermore, it is worth to work on how to further reduce parameters and FLOPs. To solve these problems, we propose a double branch network based on 3-D CNN and Transformer. The main contributions of this article are summarized as follows.

1) A multiscale spectral feature extraction (MSpeFE) module is added to the proposed network. This module consists of a channel split convolution layer with different kernel sizes on the spectral dimension, which can enrich spectral signals to improve model's performance. And similar treatment is not used in the existing hybrid models for HSI classification.

2) A TE improved by convolution operations, called ConvTE, is constructed to establish the long-distance dependencies between spectral bands. The linear layers in the TE, which include the linear projection in MHSA and linear layers in multilayer perceptron (MLP), are replaced by convolution operations to conduct convolutional multihead self-attention (CMHSA) and convolutional MLP (CMLP). In this way, not only are the number of parameters and FLOPs reduced, it also contributes to the extraction of spatial features in addition to global spectral information compared to previous Transformer-based methods. To mitigate the effects of data redundancy, convolution operations are applied to downsample the key matrix K and the value matrix V to recalibrate the spectral features.

3) A DBCT module is proposed, which contains a double branch block and a feature fusion block. Compared to the serial combination of convolution and self-attention, the parallel combination of double branch structure can model local and global representations continuously. The double branch block is developed based on a ConvTE branch and a 3-D CNN branch. The global receptive field of attention enables ConvTE branch to fully extract global spectral features. And unlike 2-D CNN, 3-D CNN branch can perform local calculations on spatial and spectral dimensions simultaneously, contributing to sufficiently extract spatial and local spectral features. Therefore, DBCT module can jointly extract spatial features and local–global spectral features, where the ConvTE branch mainly utilizes the spatial and global spectral information and 3-D CNN branch mainly utilizes the spatial and local spectral information

The rest of this paper is organized as follows. In Section II, we briefly introduce the related work. The structure of the proposed DBCTNet is detailed in Section III. The comparison experiment results of ten algorithms on four public HSI datasets, Pavia University, Houston, WHU-Hi-LongKou, and HuangHeKou, are reported in Section IV. Section V summarizes this article.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:32:34 UTC from IEEE Xplore. Restrictions apply.

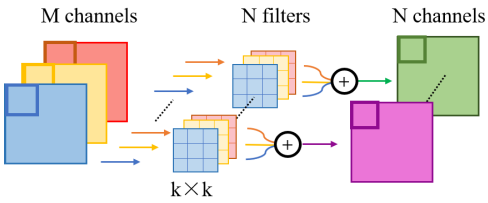


Fig. 1. Standard convolution. Every filter has M channels.

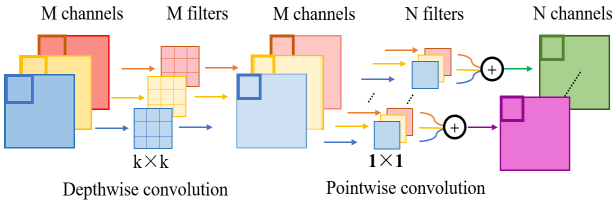


Fig. 2. DSConv. Every filter has one channel in DWConv.

II. Related Work

This section describes some basic contents related to the proposed network, including DSConv and the structure of the TE.

A. Depthwise Separable Convolution

In [54], a lightweight network was proposed based on 2-D DSConv instead of standard 2-D convolution. Standard convolution both applies each kernel to the whole input and combines the input into a set of new outputs in one step. But DSConv splits this into two layers, a depthwise convolution (DWConv) applying each kernel to each input channel separately and then a 1×1 convolution, called pointwise convolution (PWConv), combining the output of DWConv into a new output. Figs. 1 and 2 illustrate the workflow of a standard convolution and a DSConv, respectively.

The split of standard convolution into DWConv and PWConv significantly reduces the number of parameters and FLOPs and has been used for many tasks [55], [56], [57]. In the proposed work, several components are based on DSConv.

B. Structure of TE

ViT developed in [35] is an image classification model based on Transformer structure. Fig. 3 depicts the TE structure of ViT. The TE consists of multihead self-attention, MLP, layer normalization, and skip connection. Fig. 4 illustrates the multihead attention mechanism, where query, key, and value vectors are packed together into matrices Q, K, and V, respectively. MatMul refers to the matrix multiplication. For the self-attention, Q, K, and V matrices are generated from the same input by linear projection. Attention can be computed as follows:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{1}$$

where d_k is the dimension of keys. Multihead attention with h heads can be computed as follows:

$$\begin{aligned}\text{MultiHead}(Q, K, V) &= \text{Concat}(\text{head}_1, \ldots, \text{head}_h)W^O \\ \text{where head}_i &= \text{Attention}\left(QW_i^Q, KW_i^K, VW_i^V\right).\end{aligned} \tag{2}$$

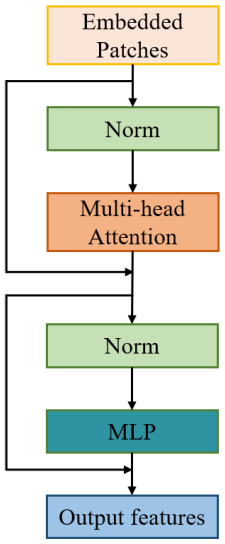


Fig. 3. Structure of the TE. MLP: multilayer perceptron. Norm: layer normalization used in TE of ViT.

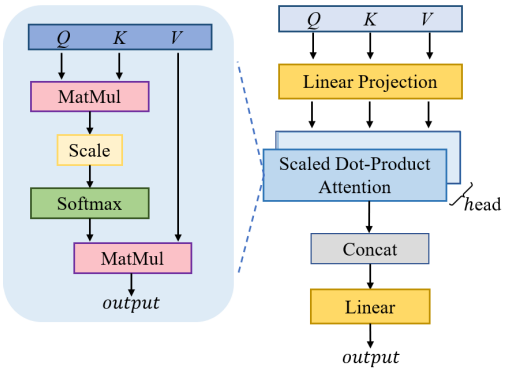


Fig. 4. Workflow of Multihead Attention. Q, K, and V represent query, key, and value matrices, respectively. MatMul means matrix multiplication.

Here, W_i^Q, W_i^K, W_i^V are parameter matrices to project Q, K, V matrices and W^O is the parameter matrix to generate output.

Due to the global receptive field of attention, the Transformer is able to powerfully extract global contextual information and model long-distance dependencies [51].

III. Proposed Network

An overview of the proposed DBCTNet is shown in Fig. 5, which mainly consists of two modules, namely MSpeFE and DBCT. MSpeFE is a multiscale spectral feature extraction module for enriching spectral signals. DBCT is a double branch module based on 3-D CNN branch and ConvTE branch to jointly extract spatial features and local–global spectral features. In the DBCT module, ConvTE can be treated as a convolution TE to extract spatial and global spectral features with a smaller number of parameters and FLOPs than standard TE.

A. Structure of the MSpeFE Module

The MSpeFE module is added at the early stage of the proposed network. A $1 \times 1 \times 7$ 3-D convolutional layer is

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:32:34 UTC from IEEE Xplore. Restrictions apply.

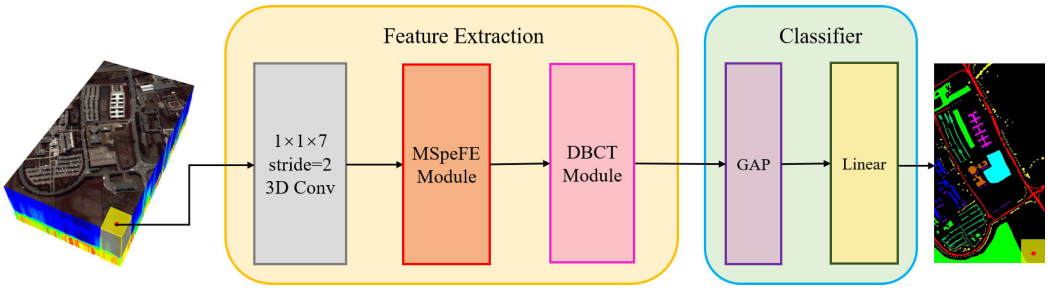


Fig. 5. Framework of the proposed DBCTNet. GAP: global average pooling.

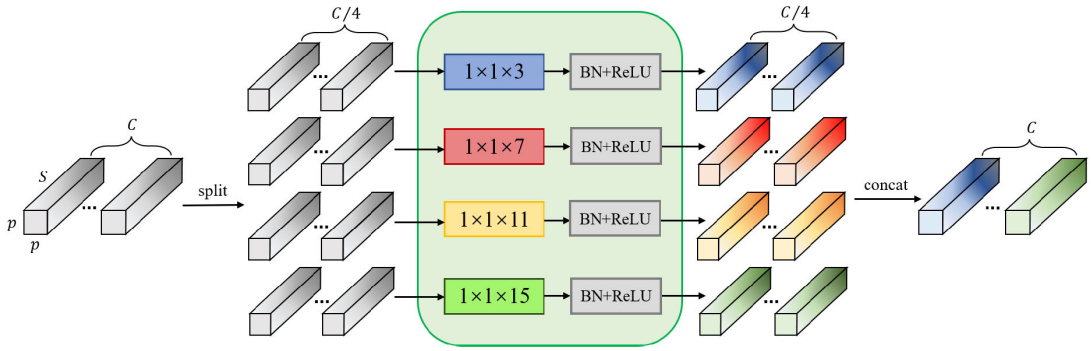


Fig. 6. Structure of the MSpeFE module. BN: batch normalization. ReLU: rectified linear unit activation function.

used to reduce the spectral dimension and then the multiscale spectral feature extraction (MSpeFE) is used. Fig. 6 shows the design of MSpeFE module, where C, p, and S denote the channel dimension, patch size, and spectral dimension, respectively.

The input is first split into four parts along the channel dimension. Then, 3-D DWConv with different convolution kernel sizes are applied to each of them. The ReLU function is placed after each convolution layer to extract spectral features at different scales. Finally, the multiscale spectral features are concatenated in the channel dimension to get the feature maps. The MSpeFE module enables the model to obtain different receptive fields to extract the spectral features at different scales, which can enrich the extracted features and improve model performance.

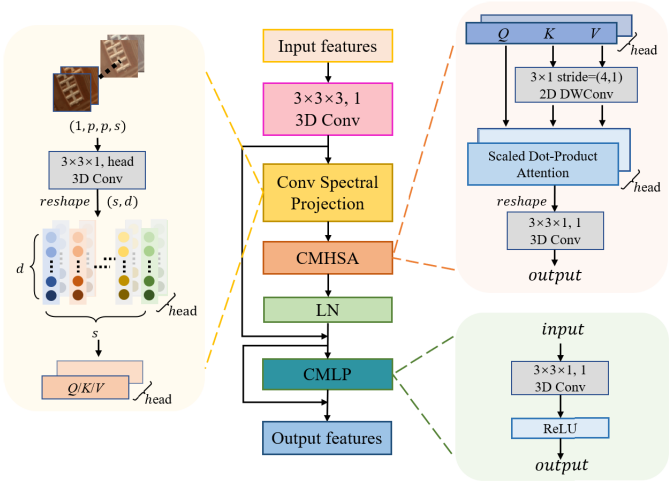


Fig. 7. Structure of the ConvTE block. LN: layer normalization.

B. Structure of the ConvTE Block

Inspired by CvT [41], we construct the ConvTE block, a generalized TE that introduces convolution operations to jointly extract spatial and global spectral features, as shown in Fig. 7. The ConvTE consists of a convolution layer, a projection layer with the help of convolution (Conv Spectral Projection), CMHSA and convolutional MLP (CMLP).

In Fig. 7, a $3 \times 3 \times 3$ convolution with one channel is used to integrate the channel information first. Next, Conv Spectral Projection is constructed by a $3 \times 3 \times 1$ 3-D convolution layer with head channels. This 3-D convolution layer can make each convolution operation only calculated within one spectral band, which means that each spectral band constitutes one token. In this way, Conv Spectral Projection can efficiently generate Q, K, and V matrices without flattening operation, which probably corrupts spatial information [50]. Then, instead of using the multihead self-attention mechanism to each spatial pixel in some existing works, such as HiT [48], CTMixer [50], CMHSA applies the multihead self-attention mechanism to each spectral band. The deployment of multihead self-attention mechanism within these bands leverages the utilization of Transformers' capacity for long-distance dependency modeling, which can capture better global spectral features from hundreds of spectral bands contained by HSI datasets. Specifically, the convolution operations are applied to downsample the K and V matrices which can recalibrate the spectral features and mitigate the effects of data redundancy.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:32:34 UTC from IEEE Xplore. Restrictions apply.

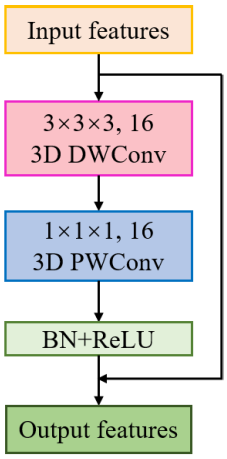


Fig. 8. Structure of the 3-D CNN branch.

Finally, CMLP, constructed by a $3 \times 3 \times 1$ 3-D convolution layer with one channel, outputs the features of ConvTE.

The TE structure allows ConvTE to extract global spectral features from the spectral bands. Compared to linear layers, convolution helps ConvTE reduce the number of parameters. Furthermore, linear projection is not good enough to capture local spatial context information [41], while Conv Spectral Projection contributes additional modeling of spatial context to enable ConvTE to extract spatial features.

C. Structure of the DBCT Module

The effectiveness of double branch networks based on 2-D CNN and TE [40], [42], [44] motivates us to further develop the Double Branch Convolution-Transformer (DBCT) module. It includes a dual branch block based on 3-D CNN and ConvTE, followed by a feature fusion block, as shown in Fig. 9.

The 3-D CNN branch is very simple in design, consisting of 3-D DWConv with a kernel size of $3 \times 3 \times 3$, 3-D PWConv, and skip connection, as shown in Fig. 8. The 3-D CNN branch can jointly extract spatial and local spectral features, while the ConvTE branch can jointly extract spatial and global spectral features. It is worth noting that the spatial features extracted by the two branches are different. The 3-D CNN branch extracts spatial features in detail because the input channels are not integrated as in the ConvTE branch. Fig. 9 depicts the structure of the DBCT module with feature fusion. With the help of 3-D DWConv, the features extracted from both branches are converted to the same size. After concatation on the channels, a 3-D PWConv is used to fuse the features.

D. Framework of the Proposed DBCTNet

The framework of DBCTNet and details on an example are given in Fig. 5 and Table I, respectively.

The HSI can be represented as a $(1, H \times W \times S)$ tensor, where 1 denotes the number of channels, H and W denote the height and width of the HSI image, respectively, and S denotes the spectral dimension. We construct patches from a HSI, which are fed into the proposed network to obtain the

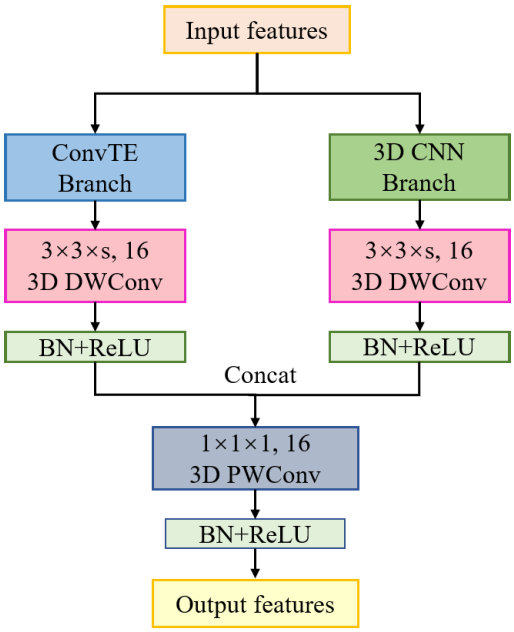


Fig. 9. Structure of the DBCT module. Concat represents tensor concatenation by the channels.

TABLE I
IMPLEMENTATION DETAILS ON AN EXAMPLE OF DBCTNET

Layer Name			Kernel Size	Input Size	Output Size
Input			-	$(1, 9 \times 9 \times 270)$	
3D Conv			$(16, 1 \times 1 \times 7)$	$(1, 9 \times 9 \times 270)$	$(16, 9 \times 9 \times 132)$
MSpeFE	Split		-	-	-
	3D Conv+BN+ReLU		$(4, 1 \times 1 \times 3)$	$(4, 9 \times 9 \times 132)$	$(4, 9 \times 9 \times 132)$
	3D Conv+BN+ReLU		$(4, 1 \times 1 \times 7)$	$(4, 9 \times 9 \times 132)$	$(4, 9 \times 9 \times 132)$
	3D Conv+BN+ReLU		$(4, 1 \times 1 \times 11)$	$(4, 9 \times 9 \times 132)$	$(4, 9 \times 9 \times 132)$
	3D Conv+BN+ReLU		$(4, 1 \times 1 \times 15)$	$(4, 9 \times 9 \times 132)$	$(4, 9 \times 9 \times 132)$
	Concat		-	-	$(16, 9 \times 9 \times 132)$
DBCT	3D CNN branch	3D DWConv	$(16, 3 \times 3 \times 3)$	$(16, 9 \times 9 \times 132)$	$(16, 9 \times 9 \times 132)$
		3D PWConv+BN+ReLU	$(16, 1 \times 1 \times 1)$	$(16, 9 \times 9 \times 132)$	$(16, 9 \times 9 \times 132)$
	ConvTE branch	3D Conv	$(1, 3 \times 3 \times 3)$	$(16, 9 \times 9 \times 132)$	$(1, 9 \times 9 \times 132)$
		Conv Spectral Projection / 3D Conv Reshape	$(2, 3 \times 3 \times 1)$	$(1, 9 \times 9 \times 132)$	$(2, 9 \times 9 \times 132)$
			-	$(2, 9 \times 9 \times 132)$	$(2, 132 \times 81)$
		CMHSA+LN	-	$(2, 132 \times 81)$	$(1, 9 \times 9 \times 132)$
		CMLP	-	$(1, 9 \times 9 \times 132)$	$(1, 9 \times 9 \times 132)$
	3D DWConv+BN+ReLU		$(16, 3 \times 3 \times 132)$	$(16, 9 \times 9 \times 132)$	$(16, 9 \times 9 \times 1)$
	3D DWConv+BN+ReLU		$(16, 3 \times 3 \times 132)$	$(1, 9 \times 9 \times 132)$	$(16, 9 \times 9 \times 1)$
	Concat		-	-	$(32, 9 \times 9 \times 1)$
	3D PWConv+BN+ReLU		$(16, 1 \times 1 \times 1)$	$(32, 9 \times 9 \times 1)$	$(16, 9 \times 9 \times 1)$
Classifier	GAP		-	$(16, 9 \times 9 \times 1)$	(16×1)
	Linear		-	(16×1)	$(n \times 1)$

classification results. For each pixel of HSI, we intercept a 3-D data cube of shape $(1, p \times p \times S)$ centered on that pixel as a sample, where p denotes the patch size. For example, when the input is a 3-D cube of shape $(1, 9 \times 9 \times 270)$, it goes through a $(1 \times 1 \times 7)$ 3-D convolution layer with 16 channels and stide $= 2$ first and then is fed into the MSpeFE module to extract the multiscale spectral features. The MSpeFE module outputs a feature map of size $(16, 9 \times 9 \times 132)$. Then, it goes through the DBCT module for the joint extraction of spatial and spectral features. For the ConvTE branch, a $3 \times 3 \times 3$ 3-D convolution with 1 channel integrates the feature maps by the channels. After the CMSHA with two heads and the CMLP layer, the ConvTE branch outputs the feature map of shape $(1, 9 \times 9 \times 132)$. For the 3-D CNN branch, the size of output is $(16, 9 \times 9 \times 132)$, which is the same as the input feature map. In the feature fusion block, two 16 channels $3 \times 3 \times 132$ 3-D DWConv layers convert the outputs of two branches into the feature maps of shape $(16, 9 \times 9 \times 1)$, respectively. After concatenating the outputs by the channels,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:32:34 UTC from IEEE Xplore. Restrictions apply.

a 16-channel 3-D PW convolution layer fuses the features to obtain the feature map of shape $(16, 9 \times 9 \times 1)$. Then, we use global average pooling to flatten the feature map into a vector and feed it into a linear layer to obtain the results of shape $(n \times 1)$, where n denotes the number of categories.

The proposed network draws on focal loss [58] in the training phase, which has been shown to be effective for HSI classification, especially when facing HSI data with unbalanced categories [21], [30], [59]. The formula for focal loss is as follows:

$$\mathrm{FL}(p_y, \alpha_y, \gamma, y) = -\alpha_y (1 - p_y)^{\gamma} \log(p_y) \tag{3}$$

where y denotes the ground-truth of the sample, p_y denotes the predicted probability that the sample belongs to the category y, and α_y, γ are the hyperparameters. The use of α_y alleviates the category imbalance problem, while the use of γ allows the network to focus on difficult samples during training.

IV. Experiment Results

In this section, we perform experiments on four public HSI datasets to verify the effectiveness of the proposed DBCT-Net. Nine DL models are selected for comparison, four of which are entirely based on CNN (RDACN [26], SSRN [27], A2S2KResNet [29], LiteDepthwiseNet [30]), three models are built based on Transformer (SpectralFormer [46], HiT [48], morphFormer [49]), and two are based on CNN combined with Transformer (CTMixer [50], HybridFormer [51]). Three quantitative metrics, overall accuracy (OA), average accuracy (AA), and κ coefficient (Kappa), are used to evaluate the performance of each method. Furthermore, we use four metrics, the number of parameters, FLOPs, GPU memory usage, and inference time, to evaluate the model's computational efficiency.

A. Dataset Description

We use the following four open HSI datasets for the experiments, **Pavia University** (**PU**), **Houston**(**HS**), **WHU-Hi-LongKou dataset** (**LK**), and **HuangHeKou** (**HHK**).

1) **PU**: This dataset was acquired by the ROSIS sensor during a flight campaign over Pavia, northern Italy. The image size of PU dataset is 610 × 340 with 103 spectral bands, and the wavelength range is 0.4–0.86 μm. PU dataset contains 10 249 labeled samples, which can be divided into nine classes.

2) **HS**: The HS dataset is provided by the HSI Analysis group and the NSF Funded Center for Airborne Laser Mapping Center(NCALM) at the University of Houston. The image size of HS dataset is 349 × 1905. HS dataset contains 144 spectral bands in the wavelength range of 0.38–1.05 μm. The dataset can be divided into 15 classes.

3) **LK**: The LK dataset was captured by an 8-mm focal length Headwall Nano-Hyperspec imaging sensor on July 17, 2018, in Longkou, Hubei, China. The size of the imagery is 550×400 pixels with 270 spectral bands from 0.400 to 1.0 μm. The study area is a simple agricultural scene with nine land cover classes, of which six are crop species.

4) **HHK**: The HHK dataset was collected by the GF5_AHSI in the area around the Yellow River Estuary of China ("Huanghekou" in Chinese) in 2019. The image size of HHK dataset is 1185 × 1342. The dataset contains 330 spectral bands in the wavelength range of 0.390–1.029 μm (VNIR) and 1.005–2.513 μm (SWIR), but 45 spectral bands are bad. We eliminated them and used the remaining 285 bands for classification. HHK dataset contains 6471 labeled samples, which can be divided into 21 classes.

We divide the PU, HS, LK, and HHK datasets into training, validation, and testing sets. For the four datasets, about 0.7%, 5%, 0.1%, and 5% of the data are used for training, respectively. The ratio of the validation set is the same as the training set, and the rest of the data are used as the testing set. For each of the four datasets, the detailed information can be seen in Tables II, III, IV, and V, respectively.

TABLE II
SAMPLES FOR EACH CATEGORY OF TRAINING, VALIDATION, AND TESTING FOR THE **PU** DATASET

No.	Class	Total number	Training	Validation	Testing
1	Asphalt	6631	47	46	6538
2	Meadows	18649	131	130	18388
3	Gravel	2099	15	14	2070
4	Trees	3064	21	22	3021
5	Painted metal sheets	1345	9	10	1326
6	Bare Soil	5029	35	35	4959
7	Bitumen	1330	9	19	1311
8	Self-Blocking Bricks	3682	26	25	3631
9	Shadows	947	6	7	934
	Total	42776	299	299	42178

TABLE III
SAMPLES FOR EACH CATEGORY OF TRAINING, VALIDATION, AND TESTING FOR THE **HS** DATASET

No.	Class	Total number	Training	Validation	Testing
1	Healthy Grass	1251	63	62	1126
2	Stressed Grass	1254	63	62	1129
3	Synthetic Grass	697	35	35	627
4	Trees	1244	62	62	1120
5	Soil	1242	62	62	1118
6	Water	325	16	17	292
7	Residential	1268	64	63	1141
8	Commercial	1244	62	62	1120
9	Road	1252	63	62	1127
10	Highway	1227	61	62	1104
11	Railway	1235	61	62	1112
12	Parking Lot1	1233	61	62	1110
13	Parking Lot2	469	24	23	422
14	Tennis Court	428	21	22	385
15	Running Track	660	33	33	594
	Total	15029	751	751	13527

TABLE IV
SAMPLES FOR EACH CATEGORY OF TRAINING, VALIDATION, AND TESTING FOR THE **LK** DATASET

No.	Class	Total number	Training	Validation	Testing
1	Corn	34511	34	35	34442
2	Cotton	8374	8	9	8357
3	Sesame	3031	3	3	3025
4	Broad-leaf soybean	63212	63	63	63086
5	Narrow-leaf soybean	4151	4	4	4143
6	Rice	11854	12	12	11830
7	Water	67056	67	67	66922
8	Roads and houses	7124	7	7	7110
9	Mixed weed	5229	6	5	5218
	Total	204542	204	205	204133

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:32:34 UTC from IEEE Xplore. Restrictions apply.

TABLE V
SAMPLES FOR EACH CATEGORY OF TRAINING, VALIDATION, AND TESTING FOR THE **HHK** DATASET

No.	Class	Total number	Training	Validation	Testing
1	Salt marsh	393	19	20	354
2	Acquaculture	796	40	40	716
3	Mud flat	110	6	5	99
4	Rice	190	10	9	171
5	Aquatic vegetation	83	4	4	75
6	Seep sea	96	5	5	86
7	Freshwater herbaceous marsh	95	5	5	85
8	Shallow sea	211	10	11	190
9	Reed	200	10	10	180
10	Pond	936	47	47	842
11	Build up	553	27	28	498
12	Suaeda salsa	469	23	24	422
13	flood plain	361	18	18	325
14	River	240	12	12	216
15	Soybean	595	29	30	536
16	Broomcorn	454	22	23	409
17	Maize	133	7	6	120
18	Locust	377	19	19	339
19	Spartina	68	4	3	61
20	Tamarix	72	4	3	65
21	Intertidal saltwater	39	2	2	35
	Total	6471	323	324	5824

B. Experimental Setting

First, all the experiments are conducted on Ubuntu 20.04.2 LTS, CPU: AMD Ryzen 7 3700X, GPU: 3090, RAM: 32 GB, GPU memory: 24 GB, Python: 3.8.5, Torch: 1.7.1+cu111.

Second, in all the experiments, we use Adam as the optimizer, and use a warmup strategy [60] to ensure the model converges stably, which is a strategy of using less aggressive learning rates at the start of training. The warmup we used is according to the following formula:

$$\mathrm{lr}(t) = \begin{cases} \frac{1}{2} \times \eta \times \left(1 + \cos\left(\pi \times \frac{t-\omega}{T-\omega}\right)\right) & \text{if } t \in (\omega, T] \\ \eta \times \max\left(0, \frac{t}{\omega}\right) & \text{if } t \in [0, \omega) \\ 0 & \text{otherwise} \end{cases} \tag{4}$$

where η denotes the maximum learning rate that can be reached by the setting, t denotes the current training step, T denotes the total number of training steps, and ω denotes the number of warmup steps. The learning rate first increases from zero to the specified learning rate η, and then decreases to zero. In all the experiments, warmup steps account for one-tenth of the total training steps. The learning rates of PU, HS, LK, and HHK datasets are set to 0.001, 0.003, 0.001, and 0.001, respectively. The weight decay of Adam is set to 0.001.

Third, we set the patch size to 9. The γ of focal loss are, respectively, set to 0.4, 1.7, 4.5, and 1.0 for PU, HS, LK, and HHK dataset. The α is the inverse of the proportion of samples in each category to the total number of samples. All the models have been trained for 300, 500, 300, and 600 epochs for PU, HS, LK, and HHK dataset with batch size 128.

C. Classification Maps and Categorized Results

*1) Classification Maps and Categorized Results for the **PU** Dataset:* The classification results of different algorithms for the **PU** dataset are shown in Table VI, and the classification maps are shown in Fig. 10. Using only 0.7% of the training samples, DBCTNet reaches 99.092%, 98.787%, and 98.797% for OA, AA, and Kappa, which is at least 0.91%, 1.723%, and 1.324% higher than the rest of the models. The proposed DBCTNet performs best on four categories with the lowest category accuracy of 94.671%. SpectralFormer performs poorly, which may be due to its insufficient use of spatial information and the lack of training samples. RDACN, LiteDepthwiseNet, and HybridFormer achieved acceptable performance on OA with slight gaps on AA and Kappa compared with DBCTNet. In addition, DBCTNet has the least number of parameters and significantly lower FLOPs than all algorithms. Furthermore, DBCTNet is not bad in terms of inference time, being slower than RDACN, SpectralFormer, and CTMixer.

*2) Classification Maps and Categorized Results for the **HS** Dataset:* The results of DBCTNet and other nine algorithms for classification of the **HS** dataset are shown in Table VII. Ground truth and the classification maps of all algorithms are shown in Fig. 11. DBCTNet achieves the most advanced performance on OA, AA, and Kappa with 98.595%, 98.181%, and 98.481%, respectively. It can be seen that HiT and morphFormer perform better than SpectralFormer, but still have certain gaps with other algorithms. SSRN is closest to the proposed DBCTNet, but it has a higher GPU memory usage. The proposed DBCTNet has the lowest number of parameters and FLOPs, only about 22K and 37M. Although the inference time of RDACN and CTMixer are faster than DBCTNet, their OA, AA, and Kappa are lower than DBCTNet by about 0.6%, 0.76%, and 0.66% at least.

*3) Classification Maps and Categorized Results for the **LK** Dataset:* The results of DBCTNet and other nine algorithms for classification of the **LK** dataset are shown in Table VIII. Ground truth and the classification maps of all algorithms are shown in Fig. 12. Factors such as imbalanced class distribution and a paucity of training samples have hampered the efficacy of algorithms like SpectralFormer. Nevertheless, DBCTNet consistently outperforms in terms of OA, AA, and Kappa metrics, registering scores of 98.155%, 93.925%, and 97.575%, respectively. What's more, the proposed DBCTNet has the lowest number of parameters and FLOPs, only about 40K and 74M, respectively. It can be observed that DBCTNet not only presents clear boundaries with minimal noise, but also stands out as the method least prone to misclassifying Narrow-leaf soybean as Cotton or Broad-leaf soybean. For OA, LiteDepthwiseNet is closest to the proposed DBCTNet. However, it necessitates a greater GPU memory allocation and incurs a protracted inference duration. Although the inference time of RDACN and CTMixer are faster than DBCTNet, its OA, AA, and Kappa are lower than DBCTNet by about 1.21%, 5.892%, and 1.61% at least. These results indicate that DBCTNet has achieved a balance between classification accuracy and model efficiency.

*4) Classification Maps and Categorized Results for the **HHK** Dataset:* The results of different algorithms for classification of the **HHK** dataset are shown in Table IX and the classification maps of different algorithms are shown in Fig. 13. DBCTNet achieves 97.167%, 94.113%, and 96.926% on OA, AA, and Kappa with the least number of parameters and FLOPs, about 42.77K and 78.85M, respectively. Owing to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:32:34 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
CATEGORIZED RESULTS FOR THE **PU** DATASET USING 0.7% TRAINING SAMPLES

Class	Color	RDACN	SSRN	A2S2KResNet	LiteDepthwiseNet	SpectralFormer	HiT	morphFormer	HybridFormer	CTMixer	DBCTNet
1		98.685	98.363	95.289	97.843	36.831	78.526	82.059	98.027	98.899	**99.434**
2		99.973	98.559	99.967	99.723	89.189	97.02	95.231	**99.995**	96.655	99.429
3		94.686	85.604	86.522	86.473	45.700	76.184	42.271	89.275	88.019	**98.116**
4		96.094	96.260	**98.544**	97.186	71.566	88.050	87.620	96.425	94.240	94.671
5		99.698	**100.00**	**100.00**	**100.00**	43.514	99.849	81.674	99.925	**100.00**	99.925
6		99.133	99.314	97.983	100.00	7.703	73.241	45.695	98.407	**100.00**	99.879
7		99.008	93.592	98.856	92.143	8.238	66.133	51.106	94.050	87.033	**100.00**
8		88.543	98.513	95.180	97.576	26.742	76.012	57.229	94.933	93.197	**99.559**
9		97.752	**99.893**	**99.893**	97.859	77.730	80.514	96.574	**99.893**	99.036	98.073
OA ×100		98.065	97.733	97.8	98.179	58.514	86.647	79.181	98.096	96.361	**99.092**
AA ×100		97.064	96.678	96.915	96.534	45.246	81.725	71.051	96.77	95.231	**98.787**
Kappa ×100		97.434	97.001	97.083	97.587	44.75	82.136	71.954	97.473	95.201	**98.797**
Parameters		433.269K	216.537K	220.824K	30.549K	177.257K	48214.898K	226.857K	2463.156K	653.993K	**16.361K**
FLOPs		141.395M	326.821M	352.772M	98.194M	88.254M	846.083M	65.520M	156.040M	216.397M	**26.363M**
GPU Memory		1899M	2257M	2849M	3089M	1965M	2939M	2625M	**1739M**	1825M	1767M
Inference Time		**1.041s**	4.320s	4.196s	16.677s	3.083s	10.263s	6.120s	6.074s	2.473s	3.164s

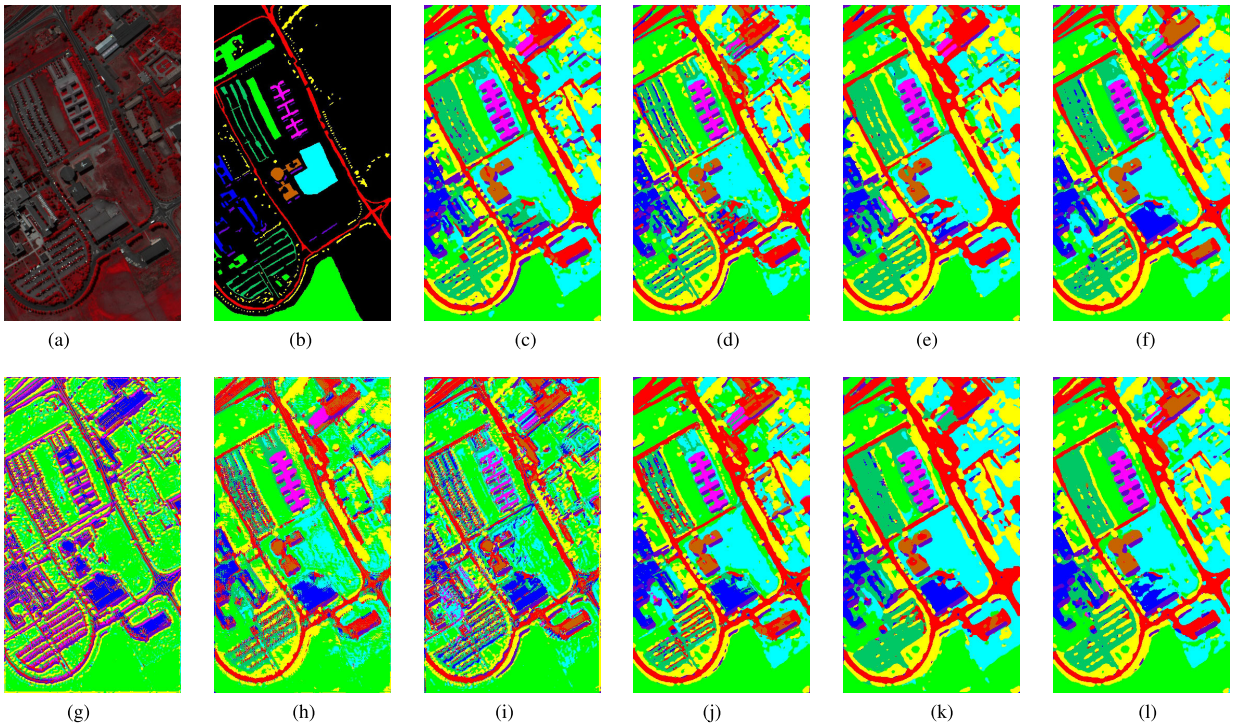


Fig. 10. Classification maps for the **PU** dataset using 0.7% of training samples. (a) False-color image. (b) GT. The classification maps with disparate algorithms (c) SSRN, (d) A2S2KResNet, (e) LiteDepthwiseNet, (f) RDACN, (g) SpectralFormer, (h) HiT, (i) morphFormer, (j) HybridFormer, (k) CTMixer, and (l) DBCTNet.

the paucity of training samples and imbalanced class distribution, the performance of SpectralFormer is inferior. HiT and morphFormer are both below 95% and 90% in OA and AA, respectively. DBCTNet outperforms HybridFormer by 1.3%, 3.26%, and 1.41% in OA, AA, and Kappa, respectively, while the inference speed is similar to HybridFormer. LiteDepthwiseNet reaches 96.446% in OA, but it has no advantage in model efficiency. SSRN, A2S2KResNet, and CTMixer achieved acceptable performance on OA with a larger number of parameters and FLOPs than DBCTNet.

D. Investigation of the Effects of the MSpeFE Module

This section discusses the effect of the MSpeFE module. We remove the MSpeFE module and compare the experiment results with original DBCTNet and the experiment settings are the same as those in Section IV-B. According to Fig. 14, it can be seen that on all four data sets, DBCTNet without MSpeFE has different degrees of decrease on OA, AA and Kappa. On average, the DBCTNet without the MSpeFE module decreases by 1.24%, 1.70%, and 1.56% on OA, AA, and Kappa, respectively. Additionally, to underscore the significance of employing varying convolution kernel sizes, we substitute MSpeFE with 3-D DWConv of size $1 \times 1 \times 3$, $1 \times 1 \times 7$, $1 \times 1 \times 11$, and $1 \times 1 \times 15$. It can be seen that the original DBCTNet achieves the best performance on all four datasets. This shows the effectiveness of the MSpeFE module where extracting multiscale spectral features may help to enrich the feature information and improve model performance.

E. Investigation of the ConvTE

To verify the rationality of Conv Spectral Projection and the downsampling operation applied to K, V matrices in the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:32:34 UTC from IEEE Xplore. Restrictions apply.

TABLE VII
CATEGORIZED RESULTS FOR THE **HS** DATASET USING 5% TRAINING SAMPLES

Class	Color	RDACN	SSRN	A2S2KResNet	LiteDepthwiseNet	SpectralFormer	HiT	morphFormer	HybridFormer	CTMixer	DBCTNet
1		97.957	99.822	99.201	**100.00**	77.265	96.359	97.336	98.490	99.467	99.290
2		99.646	99.114	98.849	98.671	57.839	98.583	97.609	99.291	**99.734**	98.937
3		**100.00**	99.841	**100.00**	99.841	87.008	98.565	96.332	**100.00**	99.841	**100.00**
4		99.464	**100.00**	**100.00**	**100.00**	43.304	97.857	92.411	99.107	99.375	99.911
5		99.911	**100.00**	**100.00**	**100.00**	86.583	96.959	94.275	**100.00**	**100.00**	**100.00**
6		84.932	84.932	**86.986**	86.301	20.89	69.863	80.137	84.247	84.932	**86.986**
7		94.829	97.458	91.937	95.706	56.091	85.276	87.642	89.746	95.793	**98.773**
8		91.250	**99.107**	89.911	89.196	40.179	83.661	82.054	95.625	98.393	96.875
9		93.256	94.055	97.604	**98.137**	37.711	86.513	73.026	91.393	94.055	96.007
10		**99.909**	98.370	97.101	99.275	55.435	86.594	80.978	95.924	99.728	99.728
11		**99.910**	98.201	96.942	97.842	50.000	76.799	81.385	96.853	96.043	99.371
12		98.829	**99.730**	98.919	99.189	56.937	87.207	88.468	97.117	99.279	98.018
13		93.365	95.261	98.104	93.365	18.483	45.261	67.773	86.967	94.550	**98.815**
14		**100.00**	**100.00**	**100.00**	**100.00**	20.26	92.468	91.948	**100.00**	**100.00**	**100.00**
15		**100.00**	**100.00**	**100.00**	**100.00**	32.828	99.663	97.811	**100.00**	**100.00**	**100.00**
OA ×100		97.383	98.344	97.206	97.664	53.589	88.726	87.743	96.222	97.989	**98.595**
AA ×100		96.883	97.726	97.037	97.168	49.392	86.775	87.279	95.651	97.413	**98.181**
Kappa ×100		97.171	98.209	96.979	97.474	49.695	87.799	86.750	95.915	97.826	**98.481**
Parameters		460.491K	278.127K	283.443K	39.555K	241.648K	51.170M	232.687K	2.778M	657.327K	**22.223K**
FLOPs		150.006M	456.214M	492.747M	137.853M	147.397M	1.074G	73.423M	256.092M	217.230M	**37.471M**
GPU Memory		1893M	2565M	3247M	3599M	2349M	3101M	2637M	**1787M**	1825M	1921M
Inference Time		**0.587s**	1.305s	2.711s	10.999s	1.286s	3.836s	2.244s	2.004s	0.840s	1.263s

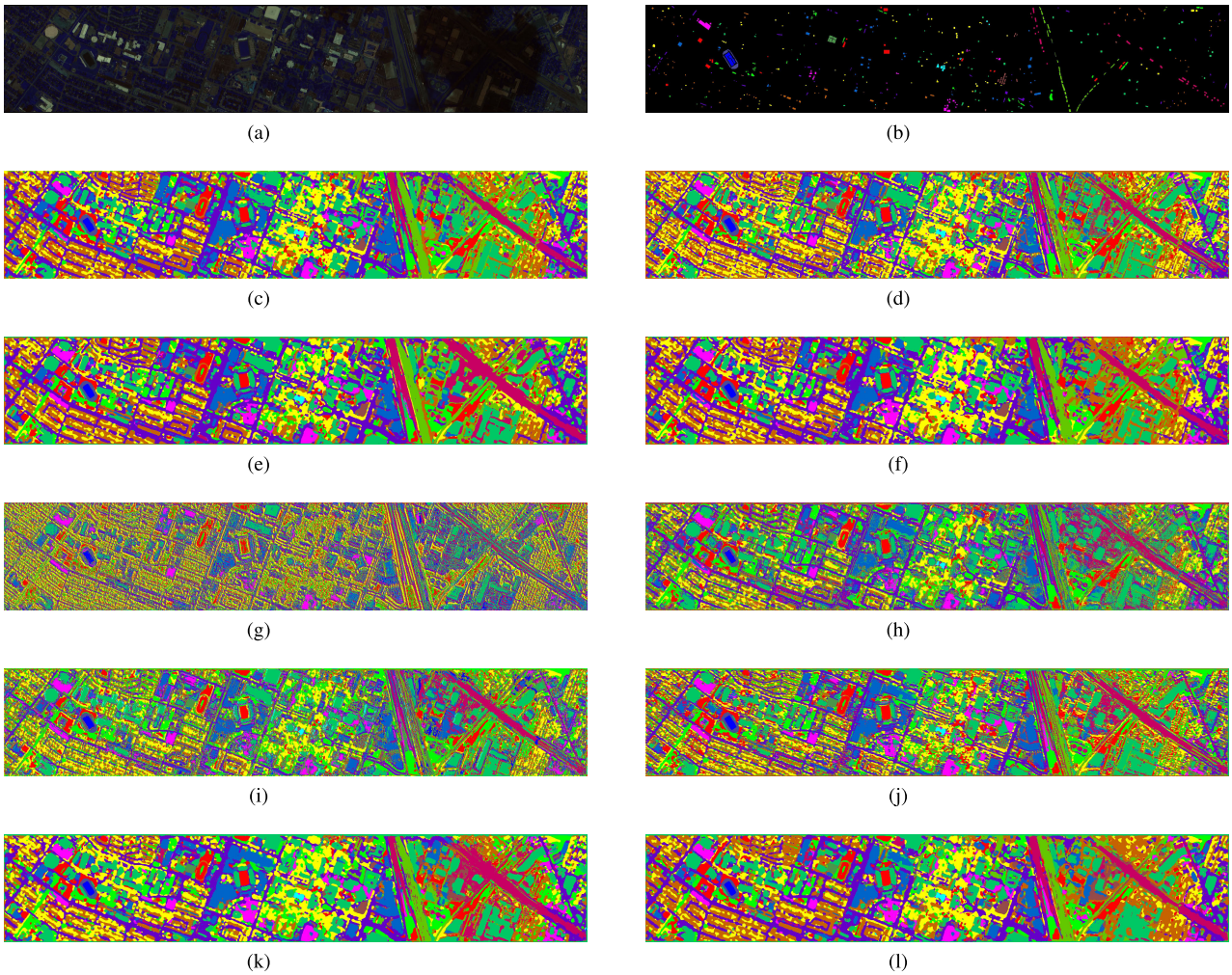


Fig. 11. Classification maps for the **HS** dataset using 5% of training samples. (a) False-color image. (b) GT. The classification maps with disparate algorithms (c) SSRN, (d) A2S2KResNet, (e) RDACN, (f) LiteDepthwiseNet, (g) SpectralFormer, (h) HiT, (i) morphFormer, (j) HybridFormer, (k) CTMixer, and (l) DBCTNet.

ConvTE block, we perform some ablation experiments on the model.

First, we compare the performance of DBCTNet when employing Conv Spectral Projection against that when using

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:32:34 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII
CATEGORIZED RESULTS FOR THE **LK** DATASET USING 0.1% TRAINING SAMPLES

Class	Color	RDACN	SSRN	A2S2KResNet	LiteDepthwiseNet	SpectralFormer	HiT	morphFormer	HybridFormer	CTMixer	DBCTNet
1		98.923	95.337	97.210	99.184	85.593	90.721	94.309	95.067	97.645	**99.289**
2		82.566	96.003	97.691	**99.067**	24.459	86.885	52.782	93.909	97.583	97.798
3		**92.033**	79.438	75.603	82.446	6.876	10.215	2.215	88.397	82.050	82.083
4		**99.645**	97.507	99.084	98.534	75.367	82.785	94.068	98.402	99.520	99.060
5		53.657	52.619	61.863	63.529	28.120	30.727	36.881	33.502	50.398	**79.339**
6		**99.383**	91.488	94.818	91.530	27.236	82.806	83.863	95.249	93.753	96.805
7		99.801	99.239	99.777	99.782	97.854	98.954	98.660	**99.985**	99.964	99.334
8		**97.032**	92.180	90.056	93.333	59.522	59.944	76.582	92.996	95.865	95.513
9		69.260	85.550	93.982	95.611	67.037	73.457	65.121	69.682	70.813	**96.110**
OA ×100		96.946	95.628	97.143	97.464	76.852	86.428	89.462	95.604	96.819	**98.155**
AA ×100		88.033	87.707	90.009	91.446	52.451	68.499	67.165	85.243	87.510	**93.925**
Kappa ×100		95.965	94.256	96.236	96.666	69.858	82.396	86.106	94.197	95.805	**97.575**
Parameters		541.485K	471.513K	479.916K	66.405K	564.196K	66632.990K	333.097K	4350.590K	622.162K	**40.265K**
FLOPs		176.457M	863.801M	933.664M	262.773M	415.425M	2287.555M	134.168M	768.001M	206.309M	**74.019M**
GPU Memory		2015M	3707M	4865M	5633M	4017M	3731M	2741M	1933M	**1879M**	2365M
Inference Time		**9.722s**	32.019s	48.340s	118.034s	29.691s	48.363s	50.599s	31.955s	14.898s	31.447s

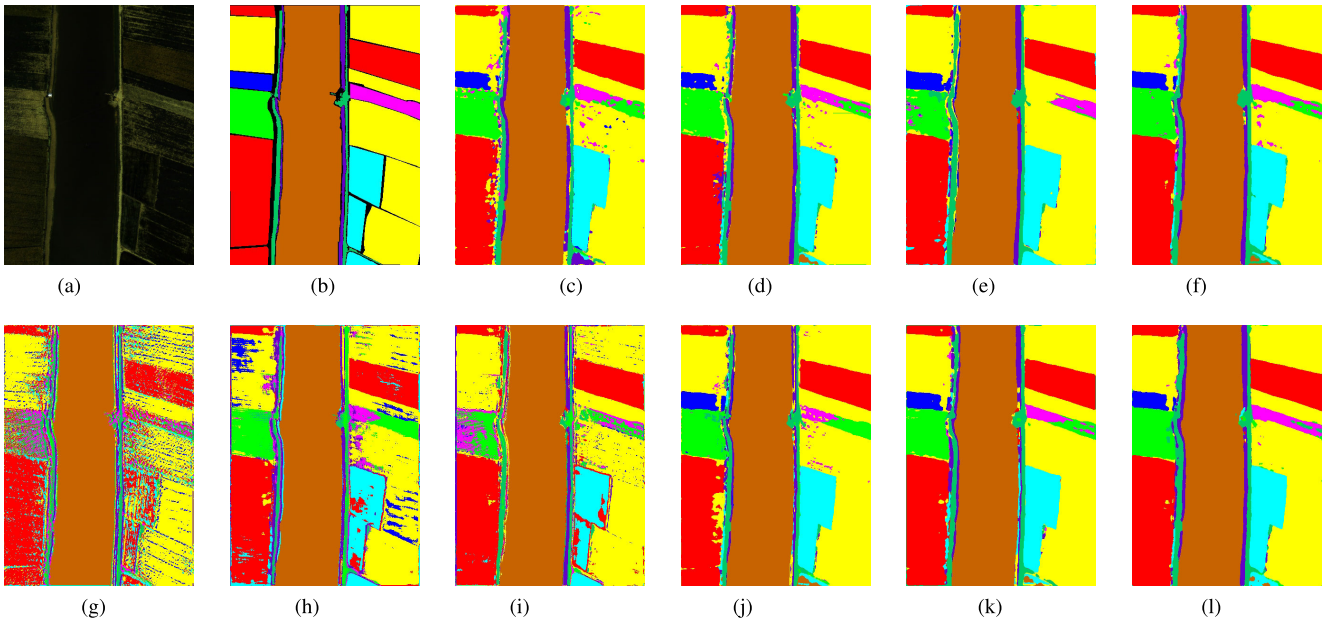


Fig. 12. Classification maps for the **LK** dataset using 0.7% of training samples. (a) False-color image. (b) GT. The classification maps with disparate algorithms (c) SSRN, (d) A2S2KResNet, (e) RDACN, (f) LiteDepthwiseNet, (g) SpectralFormer, (h) HiT, (i) morphFormer, (j) HybridFormer, (k) CTMixer, and (l) DBCTNet.

linear projection in ConvTE branch. Table X shows the results. It can be seen that the use of Conv Spectral Projection not only improve model's performance, but also reduce the number of parameters and FLOPs, which indicates that the effectiveness of replacing linear layers with convolution layers in ConvTE.

Second, we remove the downsampling operation based on convolution while performing CMHSA, and the results can be seen in Fig. 15. After removing the downsampling operation, the OA of the model is 98.307%, 98.019%, 97.137%, and 95.192% on the four datasets of PU, HS, LK, and HHK, respectively. Its OA is only lower than A2S2KResNet and LiteDepthwiseNet on the LK dataset, while the OA is still higher than all nine algorithms on the PU and HS datasets. Without downsampling, the model still performs competitively, but several gaps remain with the original DBCTNet, which indicates that the downsampling operation applied to K and V matrices for spectral recalibration may be effective.

F. Investigation of the Effects of the DBCT Module

To validate the efficacy of the DBCT module, we conducted experiments with various configurations of DBCTNet: single-branch setups with using solely the ConvTE branch or solely the CNN branch and double-branch setups using two CNN branches or two ConvTE branches. These were then benchmarked against the original DBCTNet. The results of the accuracy metrics are shown in Fig. 16, indicating that the accuracy of DBCTNet with the CNN branch removed or the ConvTE branch removed is lower than that of the original DBCTNet. Specifically, the original DBCTNet surpasses the ConvTE-only variant by an average of 1.503% in OA, 2.635% in AA, and 1.874% in Kappa. Similarly, it exceeds the CNN-only variant by margins of 0.968%, 2.413%, and 1.221% for OA, AA, and Kappa, respectively. These disparities underscore the double branch structure's importance. Moreover, when compared with double-branch setups, the original DBCTNet

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:32:34 UTC from IEEE Xplore. Restrictions apply.

TABLE IX
CATEGORIZED RESULTS FOR THE **HHK** DATASET USING 5% TRAINING SAMPLES

Class	Color	RDACN	SSRN	A2S2KResNet	LiteDepthwiseNet	SpectralFormer	HiT	morphFormer	HybridFormer	CTMixer	DBCTNet
1		**95.198**	85.311	84.463	82.486	54.237	75.989	81.638	89.266	85.028	88.418
2		98.883	**100.00**	99.581	**100.00**	98.883	99.860	99.162	99.721	**100.00**	**100.00**
3		69.697	72.727	68.687	93.939	39.394	92.929	90.909	70.707	**95.960**	86.869
4		96.491	**98.246**	**98.246**	95.322	50.877	92.398	84.795	95.906	91.228	97.661
5		77.333	**97.333**	82.667	89.333	8.000	45.333	17.333	77.333	89.333	89.333
6		98.837	97.674	89.535	81.395	20.930	90.698	47.674	65.116	72.093	**100.00**
7		95.294	85.882	95.294	92.941	25.882	68.235	40.000	88.235	**100.00**	88.235
8		**100.00**	95.789	**100.00**	**100.00**	55.263	93.158	80.000	98.947	**100.00**	95.789
9		**100.00**	**100.00**	**100.00**	**100.00**	36.667	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
10		**100.00**	97.743	97.743	**100.00**	95.606	93.824	97.031	97.268	97.506	**100.00**
11		**100.00**	**100.00**	**100.00**	**100.00**	92.972	**100.00**	99.197	**100.00**	**100.00**	**100.00**
12		**100.00**	**100.00**	**100.00**	**100.00**	99.052	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
13		99.077	**100.00**	**100.00**	**100.00**	87.692	**100.00**	96.615	**100.00**	**100.00**	**100.00**
14		**97.685**	88.426	93.519	86.574	40.278	93.056	87.963	94.444	91.204	90.278
15		**100.00**	98.507	**100.00**	**100.00**	94.963	99.254	98.507	98.507	**100.00**	**100.00**
16		**100.00**	**100.00**	98.044	98.533	86.308	**100.00**	94.866	98.533	**100.00**	98.533
17		69.167	69.167	73.333	69.167	20.833	66.667	70.000	**80.833**	74.167	72.500
18		90.265	99.115	**100.00**	98.820	53.687	87.021	91.150	97.935	92.035	99.410
19		63.934	72.131	81.967	68.852	57.377	70.492	86.885	65.574	85.246	77.049
20		90.769	90.769	72.308	90.769	33.846	38.462	47.692	**92.308**	90.769	**92.308**
21		**100.00**	**100.00**	**100.00**	**100.00**	25.714	17.143	54.286	97.143	**100.00**	**100.00**
OA ×100		96.411	96.205	96.223	96.446	76.168	92.497	91.054	95.862	96.274	**97.167**
AA ×100		92.269	92.801	92.161	92.768	56.117	82.120	79.319	90.847	93.551	**94.113**
Kappa ×100		96.106	95.885	95.904	96.142	74.054	91.853	90.293	95.511	95.959	**96.926**
Parameters		552.513K	496.389K	505.135K	70.593K	616.111K	69217.570K	425.653K	4614.044K	624.117K	**42.773K**
FLOPs		179.611M	915.559M	989.654M	278.639M	456.009M	2491.139M	167.250M	849.551M	206.448M	**78.853M**
GPU Memory		2015M	3851M	5339M	5869M	4251M	3931M	2849M	1959M	**1879M**	2439M
Inference Time		**0.260s**	1.000s	1.661s	4.432s	0.885s	1.497s	1.416s	0.927s	0.336s	0.937s

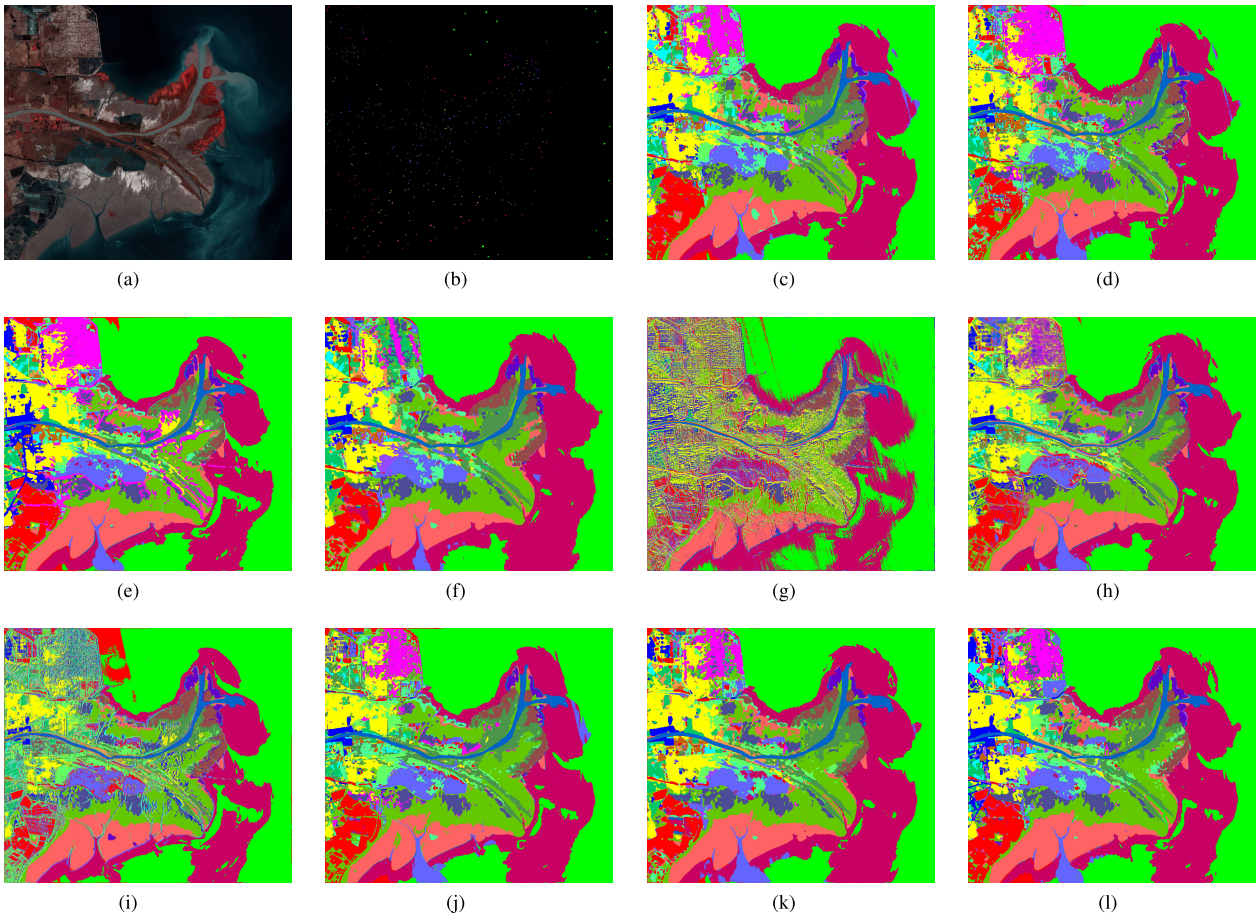


Fig. 13. Classification maps for the **HHK** dataset using 5% of training samples. (a) False-color image. (b) GT. The classification maps with disparate algorithms (c) SSRN, (d) A2S2KResNet, (e) RDACN, (f) LiteDepthwiseNet, (g) SpectralFormer, (h) HiT, (i) morphFormer, (j) HybridFormer, (k) CTMixer, and (l) DBCTNet.

still showcases superior performance, outpacing them by at least 0.604% in OA, 1.372% in AA, and 0.732% in Kappa. This contrast accentuates the advantage of combining 3-D CNN with ConvTE. Collectively, these results emphasize

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:32:34 UTC from IEEE Xplore. Restrictions apply.

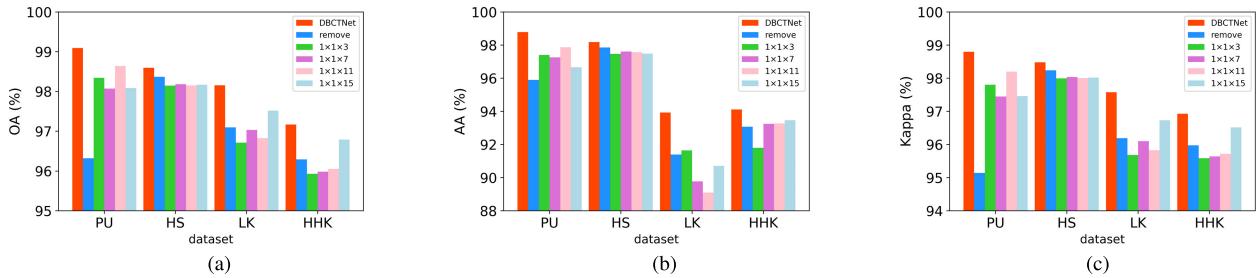


Fig. 14. Ablation study on the accuracy metrics for the MSpeFE module. (a) OA. (b) AA. (c) Kappa.

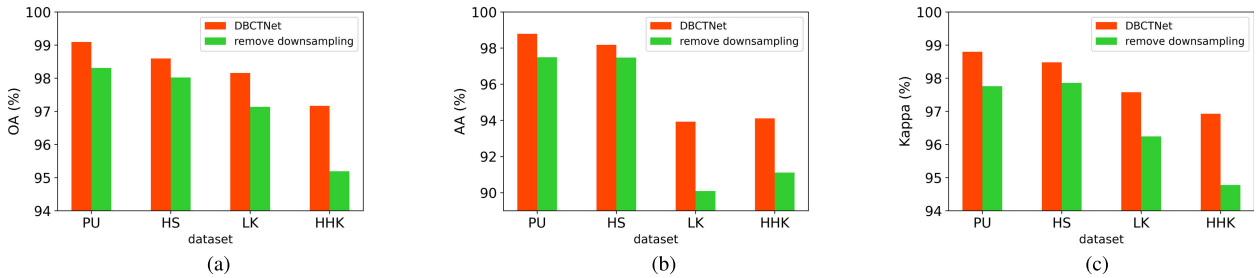


Fig. 15. Ablation study on the accuracy metrics for the downsampling applied to K and V matrices. (a) OA. (b) AA. (c) Kappa.

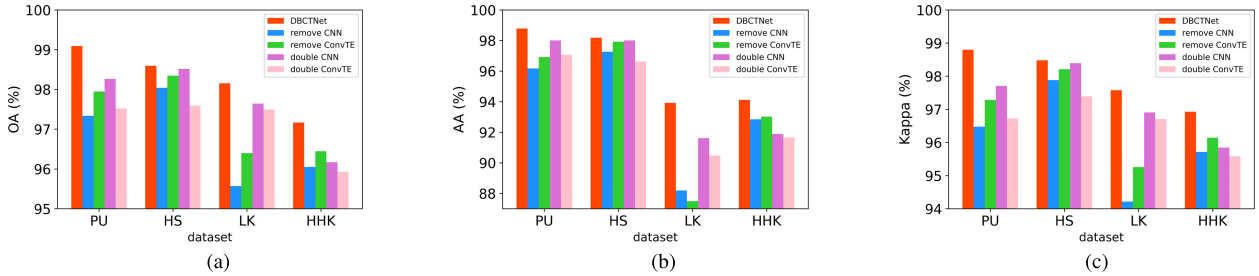


Fig. 16. Ablation study on the accuracy metrics for the DBCT module. (a) OA. (b) AA. (c) Kappa.

TABLE X
COMPARISON OF MHSA WITH CONV SPECTRAL PROJECTION AND LINEAR PROJECTION

Dataset		PU	HS	LK	HHK
Conv Spectral Projection	OA	99.092	98.595	98.155	97.167
	AA	98.787	98.181	93.925	94.113
	Kappa	98.797	98.481	97.575	96.926
	Parameters	16.361K	22.223K	40.265K	42.773K
	FLOPs	26.363M	37.471M	74.019M	78.853M
linear projection	OA	95.085	97.036	97.015	96.892
	AA	93.686	96.677	91.198	93.478
	Kappa	93.524	96.795	96.085	96.627
	Parameters	75.977K	81.839K	99.881K	102.389K
	FLOPs	36.651M	51.957M	101.732M	108.247M

the benefits of integrating CNNs with Transformers and attest to the DBCT module's effectiveness.

To further test the effectiveness of the DBCT module, we also observed the performance of these five methods for varying training sample sizes, i.e., 0.5%, 0.7%, 1.5%, and 3% for the PU dataset, 2%, 5%, 10%, and 15% for the HS dataset,0.1%, 0.2%, 0.5%, and 1% for the LK dataset, and 5%, 7%, 10%, and 15% for the PC dataset. Fig. 17 shows the OA of the five methods at varying training set sizes. It can be seen that the original DBCTNet performs best at different training sample sizes, although overall, the performance of the five methods gradually approaches as the training sample increases. These results further suggest that the excellent performance of DBCTNet may be due to the fact that the double branch structure exploits the respective advantages of 3-D CNN branch and ConvTE branch.

G. Investigation of the Effects of the Loss Function

This section explores the effect of focal loss. Proofread on all four datasets, we trained DBCTNet using standard cross entropy loss and focal loss, respectively, as shown in Table XI. It can be seen that DBCTNet with focal loss achieves better performance on all four datasets, especially on AA. Compared with the standard cross entropy loss, focal loss improves OA, AA, and Kappa by 1.224%, 6.096%, and 1.548% on average. Due to the serious class imbalance problem in the LK dataset, focal loss significantly enhanced the model's performance in terms of AA. This may be due to the fact that focal loss alleviates the category imbalance problem and focuses more on difficult samples during training. Notably, combined with the experiment results reported in Section IV-C, DBCTNet with

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:32:34 UTC from IEEE Xplore. Restrictions apply.

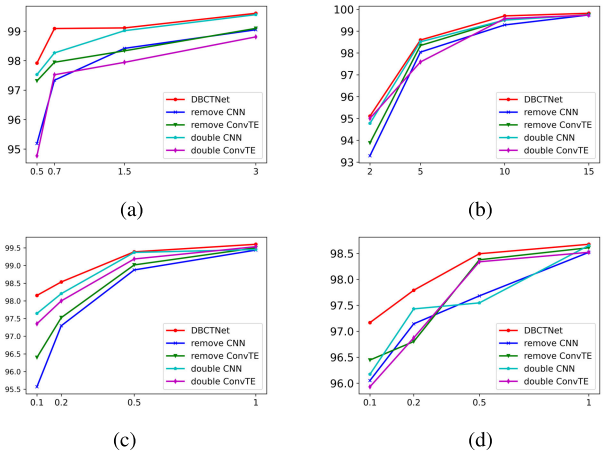


Fig. 17. Ablation study on OA for the DBCT module with different training set sizes. (a) PU. (b) HS. (c) LK. (d) HHK.

TABLE XI
COMPARISON OF DBCTNET BETWEEN DIFFERENT LOSS FUNCTIONS

Dataset	Standard Cross Entropy			Focal Loss		
	OA	AA	Kappa	OA	AA	Kappa
PU	98.430	96.484	97.917	99.092	98.787	98.797
HS	98.300	97.423	98.161	98.595	98.181	98.481
LK	95.299	74.664	93.759	98.155	93.925	97.575
HHK	96.085	92.050	95.752	97.167	94.113	96.926

standard cross entropy loss still reaches satisfactory accuracy, which illustrates the effectiveness of the DBCTNet.

V. CONCLUSION

In this article, we propose a HSI classification network called DBCTNet, which can achieve state-of-the-art performance with the help of both 3-D CNN and Transformer. First, we design the MSpeFE module to enrich feature information using multiscale spectral features. Second, we construct the ConvTE block, a generalized TE that introduces convolution operations instead of linear layers. Convolution can not only reduce the number of parameters and FLOPs, but also eliminate the need for ConvTE to flatten spatial pixels like standard TE. Furthermore, we apply downsampling to K and V matrices for spectral recalibration. Third, we establish the DBCT module, which includes a double branch block based on the 3-D CNN and ConvTE. The 3-D CNN branch can jointly extract spatial and local spectral features, while the ConvTE branch can jointly extract spatial and global spectral features. Furthermore, focal loss is used to further improve the performance of DBCTNet. Finally, we select nine existing algorithms for comparison experiments. The experiment results show that the proposed DBCTNet can achieve relatively better performance with a minimal number of parameters and FLOPs on four public HSI datasets.

In our experiments, it is observed that DBCTNet do not demonstrate notable benefits in terms of GPU Memory. This is likely attributed to its double branch design. The inference time of DBCTNet needs further improvement. In fact, the substantial memory access costs associated with DSConv can significantly extend the inference time [59], and the computational complexity of CMHSA is quadratic with respect to the number of tokens, which constrains the inference speed of DBCTNet. Additionally, the absence of an interaction module between the 3-D CNN branch and ConvTE branch, a decision made to maintain model simplicity, suggests that there is potential for further enhancing the classification performance of DBCTNet.

REFERENCES

[1] N. Yokoya, J. Chan, and K. Segl, "Potential of resolution-enhanced hyperspectral data for mineral mapping using simulated EnMAP and Sentinel-2 images," *Remote Sens.*, vol. 8, no. 3, p. 172, Feb. 2016.

[2] Y. Yuan, D. Ma, and Q. Wang, "Hyperspectral anomaly detection by graph pixel selection," *IEEE Trans. Cybern.*, vol. 46, no. 12, pp. 3123–3134, Dec. 2016.

[3] B. Tu, X. Yang, C. Zhou, D. He, and A. Plaza, "Hyperspectral anomaly detection using dual window density," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 12, pp. 8503–8517, Dec. 2020.

[4] M. Wang, Q. Wang, D. Hong, S. K. Roy, and J. Chanussot, "Learning tensor low-rank representation for hyperspectral anomaly detection," *IEEE Trans. Cybern.*, vol. 53, no. 1, pp. 679–691, Jan. 2023.

[5] C. M. Gevaert, J. Suomalainen, J. Tang, and L. Kooistra, "Generation of spectral–temporal response surfaces by combining multispectral satellite and hyperspectral UAV imagery for precision agriculture applications," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 3140–3146, Jun. 2015.

[6] A. D. Vibhute, K. V. Kale, R. K. Dhumal, and S. C. Mehrotra, "Hyperspectral imaging data atmospheric correction challenges and solutions using QUAC and FLAASH algorithms," in *Proc. Int. Conf. Man Mach. Interfacing (MAMI)*, Dec. 2015, pp. 1–6.

[7] C. Ke, "Military object detection using multiple information extracted from hyperspectral imagery," in *Proc. Int. Conf. Progr. Inform. Comput. (PIC)*, 2017, pp. 124–128.

[8] H. Yuan and Y. Y. Tang, "Spectral–spatial shared linear regression for hyperspectral image classification," *IEEE Trans. Cybern.*, vol. 47, no. 4, pp. 934–945, Apr. 2017.

[9] Y. Gu and K. Feng, "Optimized Laplacian SVM with distance metric learning for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 3, pp. 1109–1117, Jun. 2013.

[10] B. Kumar and O. Dikshit, "Spectral contextual classification of hyperspectral imagery with probabilistic relaxation labeling," *IEEE Trans. Cybern.*, vol. 47, no. 12, pp. 4380–4391, Dec. 2017.

[11] X. Sun, Q. Qu, N. M. Nasrabadi, and T. D. Tran, "Structured priors for sparse-representation-based hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 7, pp. 1235–1239, Jul. 2014.

[12] Y. Su, L. Gao, M. Jiang, A. Plaza, X. Sun, and B. Zhang, "NSCKL: Normalized spectral clustering with kernel-based learning for semisupervised hyperspectral image classification," *IEEE Trans. Cybern.*, vol. 53, no. 10, pp. 6649–6662, Oct. 2023.

[13] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Sep. 2019.

[14] W.-S. Hu, H.-C. Li, L. Pan, W. Li, R. Tao, and Q. Du, "Spatial–spectral feature extraction via deep ConvLSTM neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 4237–4250, Jun. 2020.

[15] X. Ma, H. Wang, and J. Geng, "Spectral–spatial classification of hyperspectral image based on deep auto-encoder," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 9, pp. 4073–4085, Sep. 2016.

[16] L. Gao, J. Li, K. Zheng, and X. Jia, "Enhanced autoencoders with attention-embedded degradation learning for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5509417.

[17] Z. Zhong, J. Li, D. A. Clausi, and A. Wong, "Generative adversarial networks and conditional random fields for hyperspectral image classification," *IEEE Trans. Cybern.*, vol. 50, no. 7, pp. 3318–3329, Jul. 2020.

[18] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2021.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:32:34 UTC from IEEE Xplore. Restrictions apply.

[19] Y. Dong, Q. Liu, B. Du, and L. Zhang, "Weighted feature fusion of convolutional neural network and graph attention network for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 31, pp. 1559–1572, 2022.

[20] Y. Su et al., "ACGT-net: Adaptive cuckoo refinement-based graph transfer network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5521314.

[21] Z. Lv, X.-M. Dong, J. Peng, and W. Sun, "ESSINet: Efficient spatial–spectral interaction network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5525715.

[22] P. Ghamisi, Y. Chen, and X. X. Zhu, "A self-improving convolution neural network for the classification of hyperspectral data," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 10, pp. 1537–1541, Oct. 2016.

[23] S. Mei, J. Ji, J. Hou, X. Li, and Q. Du, "Learning sensor-specific spatial–spectral features of hyperspectral images via convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 8, pp. 4520–4533, Aug. 2017.

[24] X. Wu, D. Hong, and J. Chanussot, "Convolutional neural networks for multimodal remote sensing data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5517010.

[25] M. Zhang, W. Li, Y. Zhang, R. Tao, and Q. Du, "Hyperspectral and LiDAR data classification based on structural optimization transmission," *IEEE Trans. Cybern.*, vol. 53, no. 5, pp. 3153–3164, May 2023.

[26] Z. Meng, J. Zhang, F. Zhao, H. Liu, and Z. Chang, "Residual dense asymmetric convolutional neural network for hyperspectral image classification," in *Proc. IGARSS IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 3159–3162.

[27] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral–spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2018.

[28] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3-D-2-D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Feb. 2020.

[29] S. K. Roy, S. Manna, T. Song, and L. Bruzzone, "Attention-based adaptive spectral–spatial kernel ResNet for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7831–7843, Sep. 2021.

[30] B. Cui, X.-M. Dong, Q. Zhan, J. Peng, and W. Sun, "LiteDepthwiseNet: A lightweight network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5502915.

[31] R. Song, Y. Feng, W. Cheng, Z. Mu, and X. Wang, "BS2T: Bottleneck spatial–spectral transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5532117.

[32] Y. Gao, M. Zhang, J. Wang, and W. Li, "Cross-scale mixing attention for multisource remote sensing data fusion and classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5507815.

[33] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inform. Process. Syst. (NIPS)*, 2017, pp. 5998–6008.

[34] J. Devlin, M. W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. Conf. North Amer. Chapter Assoc. Comput. Linguistics, Human Lang. Technol. (NAACL-HLT)*, 2019, pp. 4171–4186.

[35] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," in *Proc. 9th Int. Conf. Learn. Represent. (ICLR)*, 2021, pp. 1–9.

[36] Z. Liu et al., "Swin Transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9992–10002.

[37] L. Yuan et al., "Tokens-to-token ViT: Training vision transformers from scratch on ImageNet," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 538–547.

[38] W. Wang et al., "Pyramid vision transformer: A versatile backbone for dense prediction without convolutions," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 548–558.

[39] K. Wu et al., "TinyViT: Fast pretraining distillation for small vision transformers," in *Proc. Comput. Vis. (ECCV)*, 2022, pp. 68–85.

[40] Y. Chen et al., "Mobile-former: Bridging MobileNet and transformer," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 5260–5269.

[41] H. Wu et al., "CvT: Introducing convolutions to vision transformers," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 22–31.

[42] Z. Peng et al., "Conformer: Local features coupling global representations for visual recognition," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 357–366.

[43] Y. Xu, Q. Zhang, J. Zhang, and D. Tao, "ViTAE: Vision transformer advanced by exploring intrinsic inductive bias," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2021, pp. 28522–28535.

[44] Q. Chen et al., "MixFormer: Mixing features across windows and dimensions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 5239–5249.

[45] S. Mehta and M. Rastegari, "MobileViT: Light-weight, general-purpose, and mobile-friendly vision transformer," in *Proc. 10th Int. Conf. Learn. Represent. (ICLR)*, 2022, pp. 1–26.

[46] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518615.

[47] B. Tu, X. Liao, Q. Li, Y. Peng, and A. Plaza, "Local semantic feature aggregation-based transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5536115.

[48] X. Yang, W. Cao, Y. Lu, and Y. Zhou, "Hyperspectral image transformer classification networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5528715.

[49] S. K. Roy, A. Deria, C. Shah, J. M. Haut, Q. Du, and A. Plaza, "Spectral–spatial morphological attention transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5503615.

[50] J. Zhang, Z. Meng, F. Zhao, H. Liu, and Z. Chang, "Convolution transformer mixer for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[51] E. Ouyang, B. Li, W. Hu, G. Zhang, L. Zhao, and J. Wu, "When multigranularity meets spatial–spectral attention: A hybrid transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4401118.

[52] M. Jiang et al., "GraphGST: Graph generative structure-aware transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5504016.

[53] W. Liu et al., "PHTrans: Parallelly aggregating global and local representations for medical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.* Cham, Switzerland: Springer, vol. 13435, 2022, pp. 235–244.

[54] A. G. Howard et al., "MobileNets: Efficient convolutional neural networks for mobile vision applications," 2017, *arXiv:1704.04861*.

[55] K. Zhou, Y. Yang, A. Cavallaro, and T. Xiang, "Omni-scale feature learning for person re-identification," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 3701–3711.

[56] M. Ng et al., "Learning depth-guided convolutions for monocular 3D object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2020, pp. 4306–4315.

[57] Z.-X. Lan and X.-M. Dong, "MiniCrack: A simple but efficient convolutional neural network for pixel-level narrow crack detection," *Comput. Ind.*, vol. 141, Oct. 2022, Art. no. 103698.

[58] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2999–3007.

[59] S. Qiao, X.-M. Dong, J. Peng, and W. Sun, "LiteSCANet: An efficient lightweight network based on spectral and channel-wise attention for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 11655–11668, 2021.

[60] P. Goyal et al., "Accurate, large minibatch SGD: Training ImageNet in 1 hour," 2017, *arXiv:1706.02677*.



Rui Xu received the bachelor's degree from Zhejiang Gongshang University, Hangzhou, China, in 2021, where he is currently pursuing the master's degree with the School of Statistics and Mathematics.

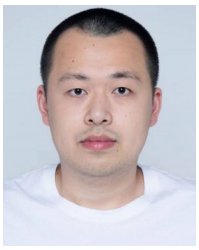
His research interests include remote sensing image processing, neural network compression, and computer vision.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:32:34 UTC from IEEE Xplore. Restrictions apply.



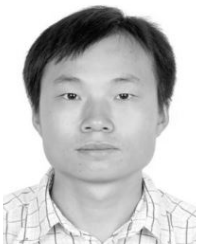
Xue-Mei Dong received the Ph.D. degree from the School of Science, Beijing University of Aeronautics and Astronautics, Beijing, China, in 2008.

From 2008 to 2010, she was a Postdoctoral Researcher at the Institute of Automation, Chinese Academy of Sciences, Beijing. She is currently a Professor with the School of Statistics and Mathematics, Zhejiang Gongshang University, Hangzhou, China. Her research interests include deep learning, remote sensing image processing, computer vision, and natural language processing.



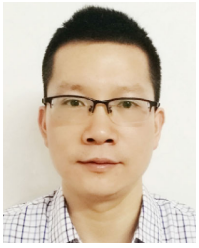
Weijie Li received the bachelor's degree from the School of Economics, Jiaxing University, Jiaxing, China, in 2020. He is currently pursuing the master's degree with the School of Statistics and Mathematics, Zhejiang Gongshang University, Hangzhou, China.

His research interests include remote sensing image processing, neural network compression, and computer vision.



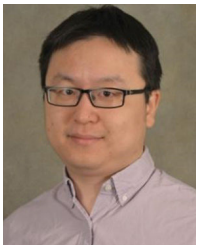
Jiangtao Peng (Senior Member, IEEE) received the B.S. and M.S. degrees from Hubei University, Wuhan, China, in 2005 and 2008, respectively, and the Ph.D. degree from the Institute of Automation, Chinese Academy of Sciences, Beijing, China, in 2011.

He is currently a Professor with the Faculty of Mathematics and Statistics, Hubei University. His research interests include machine learning and hyperspectral image processing.



Weiwei Sun (Senior Member, IEEE) received the B.S. degree in surveying and mapping and the Ph.D. degree in cartography and geographic information engineering from Tongji University, Shanghai, China, in 2007 and 2013, respectively.

From 2011 to 2012, he was a Visiting Scholar at the Department of Applied Mathematics, University of Maryland College Park, College Park, MD, USA, with Prof. J. Benedetto, where he was involved in the study of dimensionality reduction of hyperspectral image. From 2014 to 2016, he was a Postdoctoral Researcher at the State Key Laboratory for Information Engineering in Surveying, Mapping, and Remote Sensing, Wuhan University, Wuhan, China, where he was involved in the study of intelligent processing in hyperspectral imagery. From 2017 to 2018, he was a Visiting Scholar at the Department of Electrical and Computer Engineering, Mississippi State University, Starkville, MS, USA. He is currently an Associate Professor with Ningbo University, Ningbo, China. He has authored or coauthored more than 70 journal articles. His research interests include hyperspectral image processing with manifold learning, anomaly detection, and target recognition of remote sensing imagery.



Yi Xu received the Ph.D. degree in computer science from the University of Iowa, Iowa City, IA, USA, in 2019.

He is currently a Professor with the School of Control Science and Engineering, Dalian University of Technology, Dalian, China. He has published more than twenty papers in refereed journals and conference proceedings, including IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE (TPAMI), *Transactions on Machine Learning Research*, Conference on Neural Information Processing Systems (NeurIPS), International Conference on Machine Learning (ICML), and Conference on Computer Vision and Pattern Recognition (CVPR). He serves as an SPC/PC/Reviewer in these leading conferences. His research interests include machine learning, optimization, deep learning, and statistical learning theory.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:32:34 UTC from IEEE Xplore. Restrictions apply.